MA 324
Statistical Inference and Multivariate Analysis
IIT GUWAHATI

End-Semester Examination
14:00–17:00 IST
May 8, 2026

PART-B

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Instructions

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	1	3	5	1	10

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & (S_{y\boldsymbol{X}})^T \\ S_{y\boldsymbol{X}} & S_{\boldsymbol{XX}} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1\times1}$ and $\boldsymbol{X}_{q\times1}$ based on a centered (with respect to sample mean) sample of size n. Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector $\boldsymbol{X}$.

- We know canonical correlation (Corr$_i$) $\Rightarrow$ $\text{Corr}(U_i = a_i^T Y, V_i = b_i^T X)$

We maximize the Corr$_i$ ~~w.r.t~~ subject to $\text{Var}(U_i) = 1$ & $\text{Var}(V_i) = 1$

Let us look at first canonical correlation

$$\text{Corr}(U_1 = a_1^T Y, V_1 = b_1^T X) = \frac{a_1^T \Sigma_{12} b_1}{\sqrt{a_1^T \Sigma_{11} a_1}\sqrt{b_1^T \Sigma_{11} b_1}}$$

$$= \frac{a_1^T S_{yx}^T b_1}{1 \cdot 1}$$

$$= a_1^T S_{yx}^T b_1 \quad \text{— (1)}$$

~~Here a_1~~

- Given: $\text{Corr}(U_1, V_1) = r_1^2$

- Here for first canonical correlation a_1 & b_1 are ~~first~~ ~~at largest~~ eigenvector corresponding to largest eigen-value $((r_1^2)^2)$ of $(\Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1/2})$ ~~$S_{yy}^{-1/2} S_{yX}^T S_{XX}$~~

$S_{yy}^{-1/2} S_{yx}^T S_{xx}^{-1} S_{yx} S_{yy}^{-1/2}$ & $S_{xx}^{-1/2} S_{yx} S_{yy}^{-1} S_{yx}^T S_{xx}^{-1/2}$ respectively.

where $U_1 = a_1 S_{yy}^{-1/2} Y$

$V_1 = b_1 S_{xx}^{-1/2} X$

$Corr(U_1, V_1) = r_1^2$ — (2)

• lets see coeff of determination ~~of~~ (R^2) obtained from MLR of Y on X

so ~~$Y = X\beta + \varepsilon$~~
$Y = X\beta + \varepsilon$ where $\beta = [\beta_0 \ \beta_1 \ \cdots\cdots \ \beta_q]^T$

$X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1q} \\ \vdots & & & \\ 1 & x_{n1} & \cdots & x_{nq} \end{bmatrix}$

X

• we know
Coeff. of ~~Det~~ Determination $(R^2) = 1 - \dfrac{SS_{Res}}{SS_T} = \dfrac{SS_{Reg}}{SS_T}$

where $SS_{res} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

$SS_T = \sum_{i=1}^{n} (y_i - \bar{y})^2$ $SS_{Reg} = SS_T - SS_{Res}$

In our case

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}$$

• So, $r_1^2 = a_1 S_{yy}^{-1/2} S_{yx}^T b_1 S_{xx}^{-1/2}$ Using (1) & (2)

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2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.

   (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.

   (b) (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.

   (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^{p} \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

$$\underset{4\times1}{X} = \underset{4\times1}{\mu} + \underset{4\times2}{L}\,\underset{2\times1}{F} + \underset{4\times1}{\varepsilon}$$

(a) $\tilde{L} = \left[\sqrt{\lambda_3}\, e_3 \;\middle|\; \sqrt{\lambda_2}\, e_2\right]$

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$= \left[\sqrt{4.2}\, e_3 \;\middle|\; \sqrt{2.1}\, e_2\right]$

$\tilde{L} = \left[2.049\, e_3 \;\middle|\; 1.449\, e_2\right]$

where $2.049\, e_3$ & $1.449\, e_2$ are 4×1 ~~column~~ COLUMN vector acting as COLUMN 1 & 2 of $\tilde{L}_{4\times2}$ matrix respectively.

(b) We know $\tilde{h}_1^2 = l_{11}^2 + l_{12}^2$

first row of $\tilde{L} = \tilde{l}_1^T = [0.8 \quad -0.4]$

So, by comparing we get $l_{11} = 0.8$ $\quad l_{12} = -0.4$

So, $\tilde{h}_1^2 = 0.64 + 0.16 = 0.8$

$\boxed{\therefore\ \tilde{h}_1^2 = 0.8}$

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(c) $\sum_{i=1}^{p} h_i^2 = \sum_{i=1}^{p} \sum_{j=1}^{m} l_{ij}^2$

Since $X = \mu + LF + \varepsilon$ where ε is special variance matrix

L is loading factor matrix

F is factor matrix

μ is mean matrix

Take variance of ~~eq~~ above eq^n

we know

$\Sigma = LL^T + \varphi$ where $Var(X) = \Sigma$

as per our assumptions

$Var[\varepsilon] = \varphi$ a diagonal matrix

$\Rightarrow \sigma_{ii} = h_i^2 + \varphi_i$

Replace Σ by S we will get estimated ~~values~~

$\tilde{\sigma}_{ii} = \tilde{a}_i^2 + \tilde{\varphi}_i$

$\sum_{i=1}^{p} a_i^2 = \sum_{i=1}^{p} \tilde{\sigma}_{ii} - \sum_{i=1}^{p} \varphi_i = \sum_{i=1}^{p} \tilde{\sigma}_{ii}$ as $E[\varphi] = 0$

$= trace(S) = \sum_{i=1}^{p} \lambda_i$

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3. (5 points) Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2. Let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Derive a variance-stabilizing transformation $g(\overline{X}_n)$ such that the asymptotic variance of $g(\overline{X}_n)$ does not depend on μ.

• $E[X_i] = \mu$ $\quad$ $Var(X_i) = \mu^2$ $\quad$ $i = 1, 2 \ldots n$

• $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ $\quad$ $E[\bar{X}_n] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} E[X_i]$

$= \frac{1}{n} \cdot n\mu$

$\boxed{E[\bar{X}_n] = \mu}$

$Var(\bar{X}_n) = Var\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2} Var\left(\sum_{i=1}^{n} X_i\right) = \frac{n\mu^2}{n^2} = \frac{\mu^2}{n}$

• We notice that Variance of $\bar{X}_n$ depends on μ i.e.

$Var(\bar{X}_n) \propto \mu^2$

• Variance stabilizing transformation be $g(\bar{X}_n)$

$\cancel{g(\bar{X}_n) = \int \frac{1}{\sqrt{Var(\bar{X}_n)}}}$

$g(\mu) = \int \frac{1}{\sqrt{Var(\bar{X}_n)}} d\mu = \int \frac{1}{\sqrt{\mu^2}} d\mu = \log(\mu)$

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So, $g(\bar{X}_n) = \log(\bar{X}_n)$ is variance stabilizing transformation.

Proof : • Taylor formula of $g(Y)$ ~~around~~ around $E(Y)$ where $Y = \bar{X}_n$

• $g(Y) = g(E[Y]) + (Y - E[Y])\, g'(\tilde{Y})$

• Take variance of ~~both side~~ BOTH sides

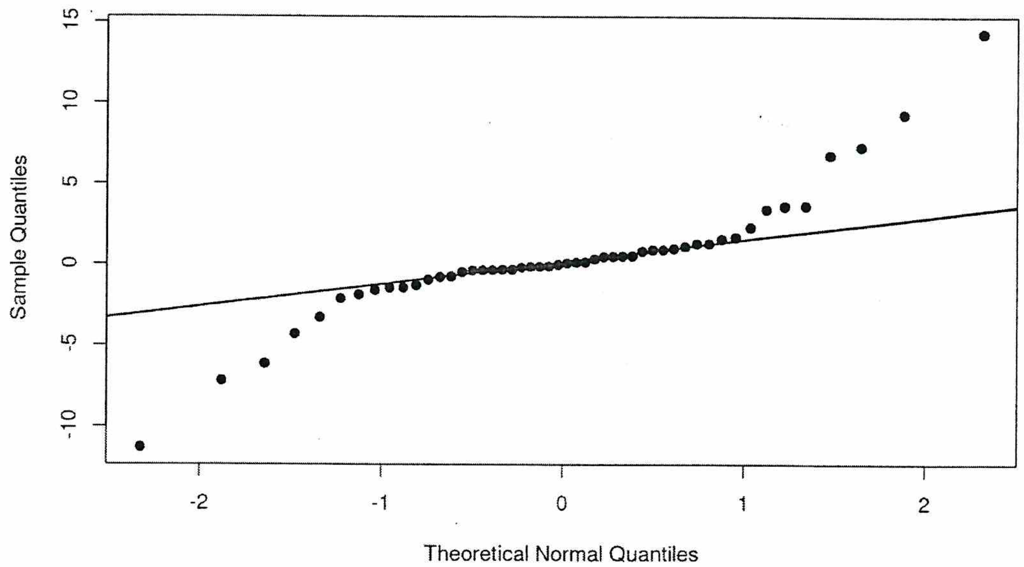
• $Var(g(Y)) = 0 + Var(Y g'(\tilde{Y})) = (g'(\tilde{Y}))^2 Var(Y)$

$g'(\tilde{Y}),\ g(E[Y]) \rightarrow$ CONSTANSTS

• We want $\rightarrow$ $Var(g(Y)) \propto 1 \Rightarrow (g'(\tilde{Y}))^2 Var(Y) \propto 1$

$\Rightarrow g'(\tilde{Y}) \propto \frac{1}{(Var(Y))^{1/2}} \Rightarrow g(\mu) = \int \frac{1}{\sqrt{Var(Y)}} d\mu$

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



No, the data does not follow a normal distribution as the Q-Q plot does not make a 45° ~~line~~ straight line & have ~~long Tail (Platykurtic) plot~~. Short Tail (leptokurtic).

X

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Additional Pages for Answers